



TEAM LEADER

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PROBLEM STATEMENT

FAIR DROP





WHY WE CHOSE THIS PROBLEM

50,000

USERS



500

SEATS



1000s

BOT REQUESTS / SEC

BOTS OUT-SEND HUMANS

Real users lose seats even when ready at the same time.

MORE REQUESTS $\neq$ FAIR

Sending more requests should never raise your chances.

TRAFFIC $\neq$ FAIRNESS

Overload causes timeouts and inconsistent allocation.

Limited-seat registration should be a fair opportunity, not a race between humans and bots.



FAIR DROP: OUR SOLUTION

1

DETECT ABUSE

Flags high frequency, repeats and abnormal speed. Rate-limit, challenge or block.

2

FAIR ALLOCATION

Eligible users enter a controlled queue. Selection is fair, not fastest-wins.

3

NO DUPLICATES

Atomic transactions and locking: one seat, one winner. Never oversold.

4

SESSION SAFE

Refresh, reconnect or reload and your reservation state survives.

5

HANDLES TRAFFIC

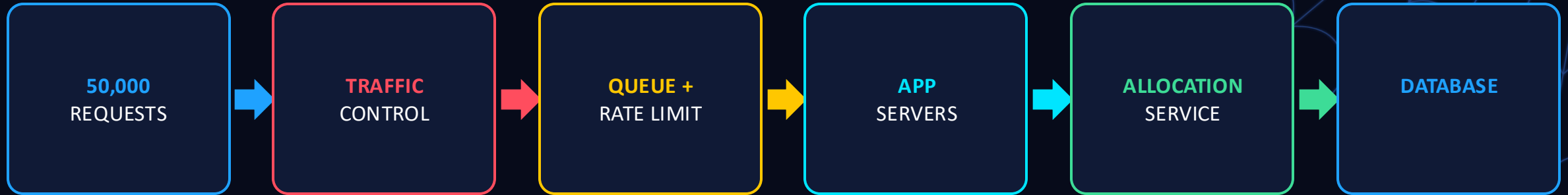
Queue and rate limiter shield the database from the flood.

We do not out-run bots. We make speed and volume irrelevant to winning.



HOW IT WORKS

TRAFFIC PIPELINE



FAIR ALLOCATION FLOW



Only eligible traffic reaches the allocation service, and the database never sees the flood.



FEATURES WE BUILT

SECURE SIGN-IN

Separate user and admin portals. Salted script passwords and server-issued 8-hour sessions.

LIVE QUEUE

Join on your own click. Live position, progress bar and wait estimate from the server.

IMAGE CAPTCHA

Server-issued, single-use and expiring. Guards sign-in, sign-up and every seat claim.

SEAT HOLD TIMER

A 2-minute reservation with a live countdown. Expired holds return to the pool.

BOT DEFENSE

Flags sub-3ms spacing, fake fingerprints, CAPTCHA replays and 50+ req/s per IP.

ADMIN CONSOLE

Launch 500 humans, a 5,000-bot horde or a flash crowd and watch the live audit log.



BUILT FOR LIVE FAIRNESS

LIVE SEAT MATRIX

All 500 seats update every second for buyers and admins. Your own seat is outlined.

YOU CONTROL THE QUEUE

Nobody is joined automatically. Logging out leaves the queue and frees any held seat.

CLEAR CAPTCHA

Six easy-to-read characters, no look-alikes, any case. Helpful errors and a fresh code each time.

ONE SEAT, ONE BUYER

The server checks your turn has arrived and blocks a second purchase. Never oversold.

SURVIVES A RELOAD

Refresh the tab and your queue spot, active hold or finished ticket are restored.

SERVER-SIDE STRESS TESTS

Surges and bot floods fill seats on the server, so every open screen sees the same grid.



SECURE SIGN-IN AND SIGN-UP

Sign In

Choose your portal, then sign in

USER

ADMIN

Email

Password

M B R B G T

Enter CAPTCHA

SIGN IN

Hello, Friend!

Create a purchaser account and open the User portal in a new tab.

SIGN UP

SIGN IN: CHOOSE USER OR ADMIN, THEN SOLVE THE CAPTCHA

Create Account

Register as a ticket purchaser

Name

Email

Password (min 6 chars)

J G K A T Z

Enter CAPTCHA

SIGN UP

Welcome Back!

Sign in to open your authenticated portal in a separate

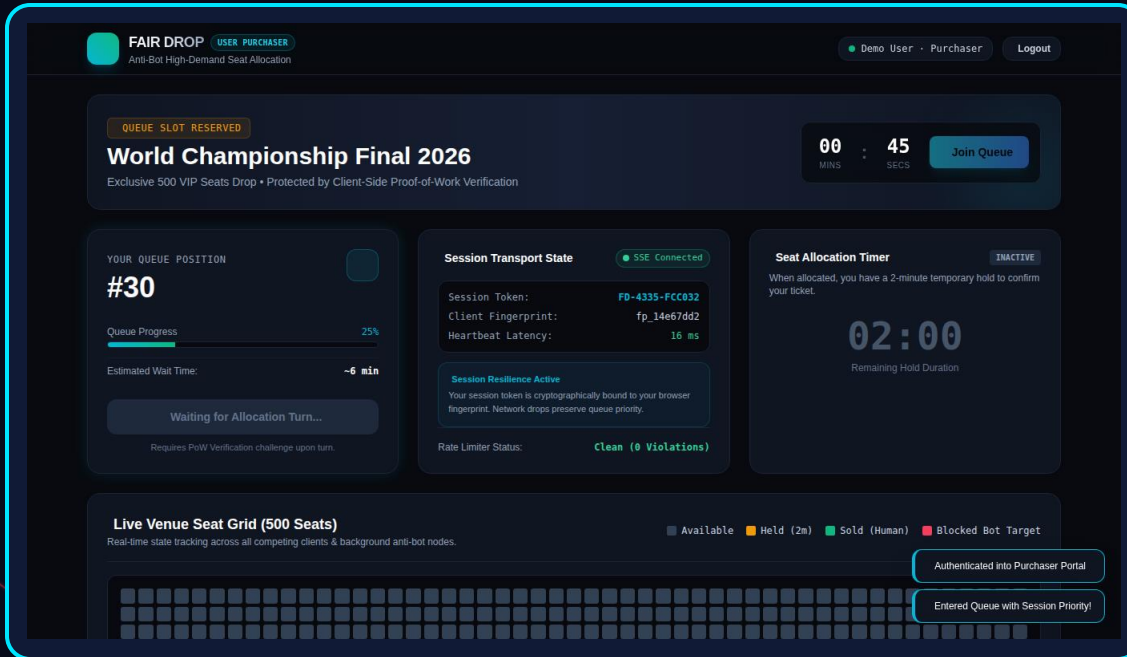
SIGN IN

SIGN UP: NEW PURCHASERS PROVE THEY ARE HUMAN TOO

No CAPTCHA, no account and no session. Bots are stopped at the front door.



LIVE BUILD: JOINING THE QUEUE



QUEUE ENTRY WITH SESSION PRIORITY

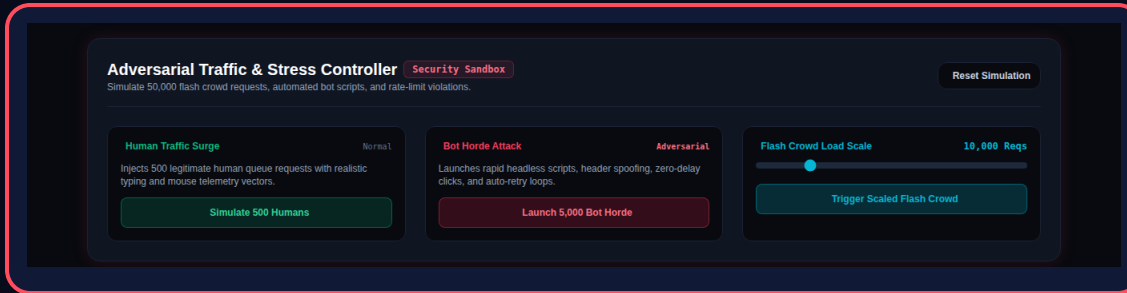
WHAT THE BUYER SEES

- Join with one click, never automatically
- Live position, progress bar and wait estimate
- Position moves as the server clears the line
- Reload the tab and your place is restored
- At READY, solve a CAPTCHA to claim a seat

You choose when to join. The server decides when your turn arrives.



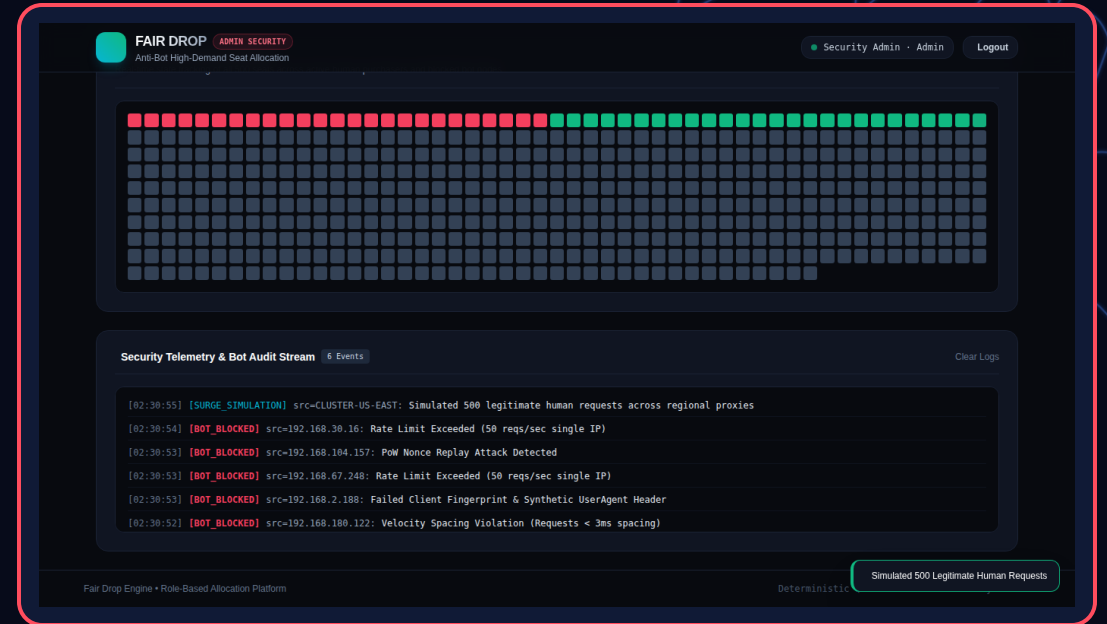
LIVE BUILD: ADMIN CONSOLE



ATTACK CONTROLS: HUMANS, BOTS, FLASH CROWD

WHY BOTS GET BLOCKED

- Request spacing under 3ms
- Synthetic fingerprint or user agent
- CAPTCHA replay attempts
- More than 50 requests/sec per IP



SEAT GRID AND LIVE BOT AUDIT STREAM



CLAIM YOUR SEAT: THE CAPTCHA

YOUR QUEUE POSITION

READY

Queue progress 100%

Estimated wait ~0 min

Verify & Claim Allocated Seat

Requires a fresh server-side CAPTCHA upon turn.



Anti-Bot Verification

CAPTCHA Seat Verification

Enter the server-issued CAPTCHA to generate a single-use seat claim.

V U 3 B N Q

V U 3 B N Q

Submit & Claim Seat

THE CAPTCHA

- Six clear characters, drawn by the server
- Type it in any case, press Enter
- A wrong answer loads a fresh code
- The right answer unlocks a 2-minute seat hold

STEP 1: YOUR TURN ARRIVES. QUEUE SHOWS READY

STEP 2: CLICK VERIFY AND CLAIM. A CAPTCHA APPEARS

Easy for a person. A bot has to solve it for every attempt.



TEMPORARY ALLOTMENT: 2-MINUTE HOLD

YOUR QUEUE POSITION

READY

Queue Progress

100%

Estimated Wait Time:

~0 min

Verify & Claim Allocated Seat

Requires PoW Verification challenge upon turn.

Session Transport State

SSE Connected

Session Token:

FD-0A3D-330D49

Client Fingerprint:

fp_020a42eb

Heartbeat Latency:

16 ms

Session Resilience Active

Your session token is cryptographically bound to your browser fingerprint. Network drops preserve queue priority.

Rate Limiter Status:

Clean (0 Violations)

Seat Allocation Timer

2M HOLD ACTIVE

When allocated, you have a 2-minute temporary hold to confirm your ticket.

01:59

Remaining Hold Duration

Selected Seat:

Seat #1

Price Tier:

\$150.00 (Standard VIP)

Confirm Purchase Now



SEAT #1 HELD (AMBER)

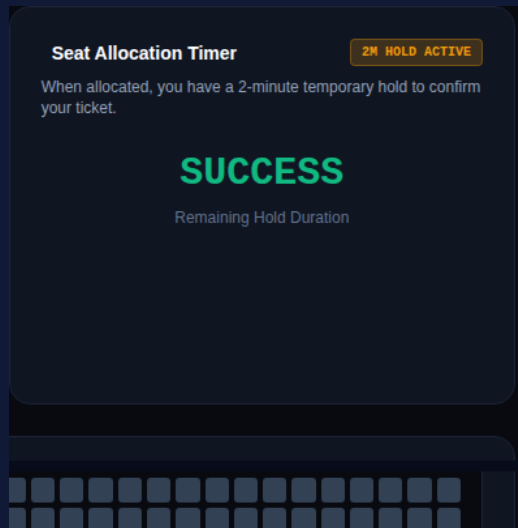
COUNTDOWN FROM 02:00

\$150 STANDARD VIP



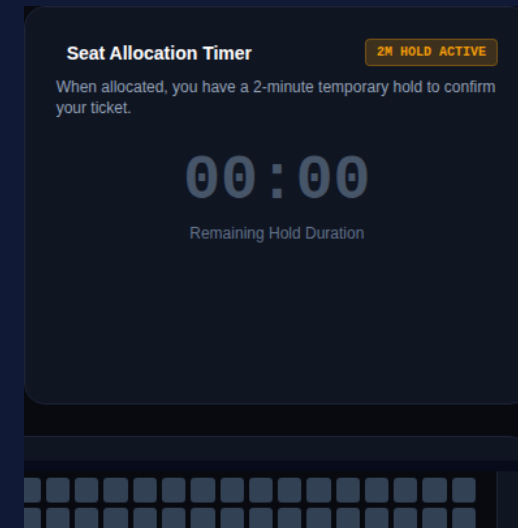
TWO SCENARIOS AFTER THE HOLD

SCENARIO 1: BUYS IN TIME



- Clicks Confirm Purchase Now
- Seat turns green (sold)
- Timer shows SUCCESS
- Ticket is claimed

SCENARIO 2: TIMER EXPIRES



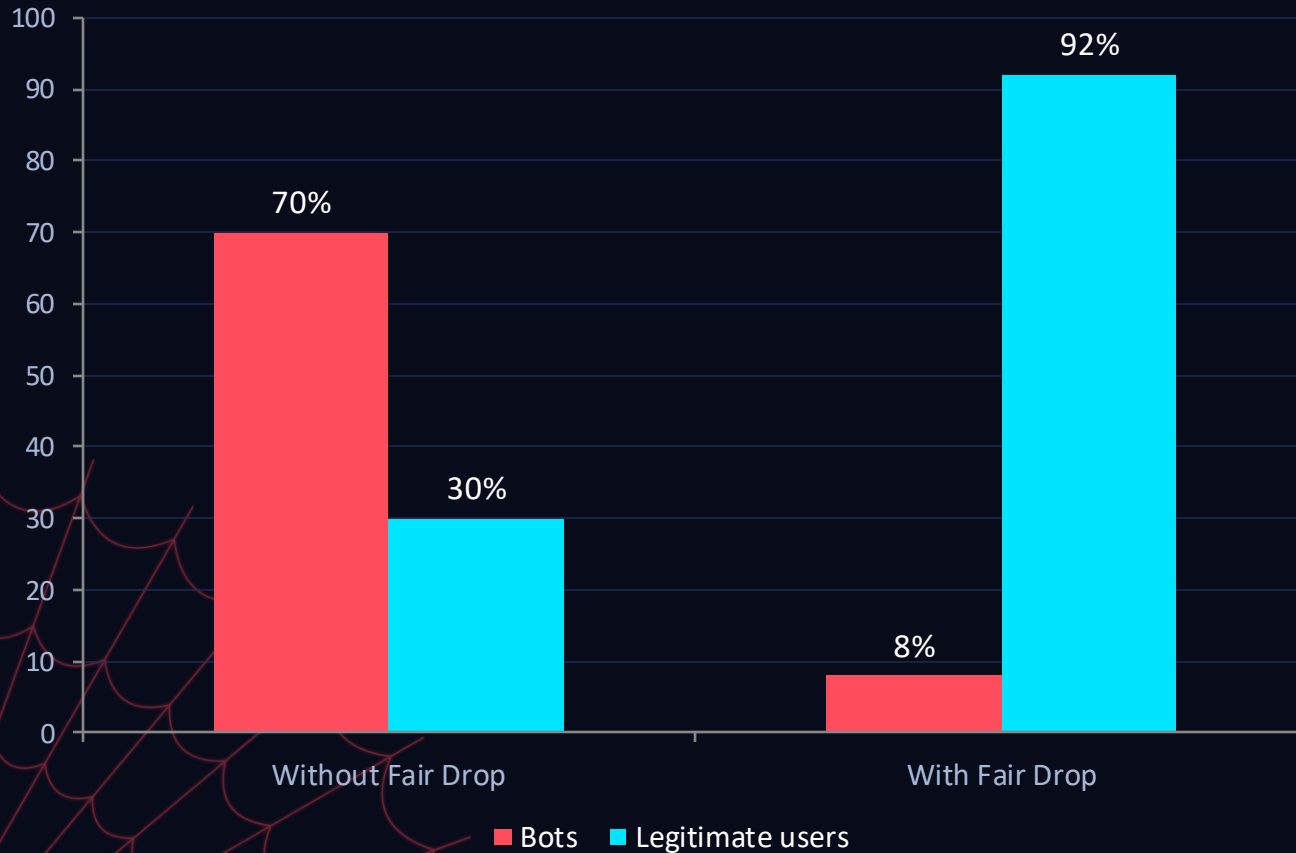
- Countdown reaches 00:00
- Seat returns to available
- Toast: reservation expired
- Next user can claim it

Held seats never block inventory for long: buy within 2 minutes or the seat goes back.



BOTS VS. FAIR DROP

SEATS WON (%) - ILLUSTRATIVE TARGET



NORMAL SYSTEM

First request wins. A bot sending 5,000 requests gets ~5,000 chances.

FAIR DROP

5,000 requests is still one fair chance. Volume buys nothing.



HOW WE PROVE IT WORKS

SIMULATED USER TYPES

USER TYPE	BEHAVIOUR
Normal	1 request
Slow user	Requests with delays
Aggressive client	Many requests
Bot	Extreme request rate
Repeat attacker	Constant retries

WHAT WE MEASURE

Allocation success rate

Bot vs legit allocation

Duplicate attempts blocked

Request throughput

Response time

Error rate

Server load

Rate-limited requests

Compare every metric with and without Fair Drop.



THE CORE MESSAGE

PROBLEM

The fastest, most aggressive client wins.

SOLUTION

Eligible users get a fair chance; abusive traffic is controlled.

GOAL

500 seats allocated fairly, not won by the fastest bot.